Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **50**

Instructor Name: **Bodanske, Daniel**

Question 1 of 50

Consider $\triangle GHJ$ in the figure below.

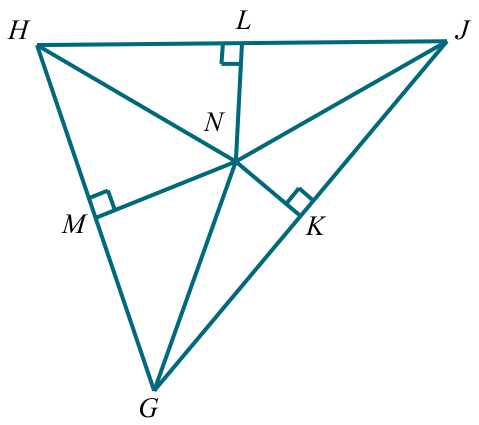
The perpendicular bisectors of its sides are $\overline{KN}$, $\overline{LN}$, and $\overline{MN}$. They meet at a single point N.

(In other words, N is the circumcenter of $\triangle GHJ$.)

Suppose $LN = 22$, $GK = 82$, and $GN = 122$.

Find HN, LJ, and GJ.

Note that the figure is not drawn to scale.



$HN =$ _____

$LJ =$ _____

$GJ =$ _____

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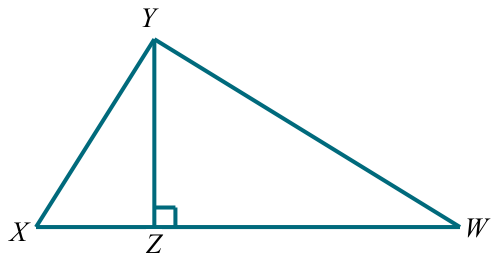
Question 2 of 50

In the figure below, there are three right trangles. Complete the following.



(a) Write a similiarity statement relating the three right triangles.

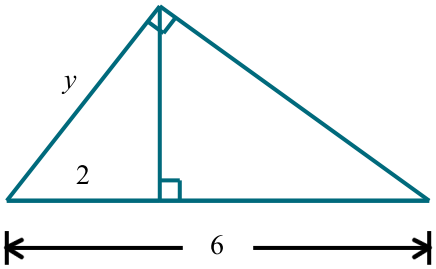
$\triangle YWZ \sim \triangle \square \sim \triangle \square$

(b) Complete each proportion.

$\frac{XZ}{XY} = \frac{XY}{\square}$ $\qquad$ $\frac{WZ}{YZ} = \frac{\square}{XZ}$

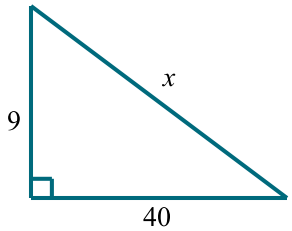
Question 3 of 50

In the figure below, find the exact value of y. (Do not approximate your answer.)



Question 4 of 50

For the following right triangle, find the side length x.



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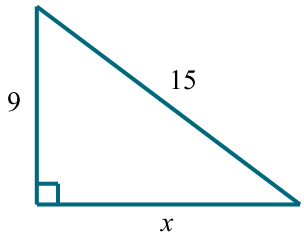
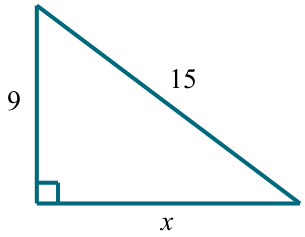
Question 5 of 50

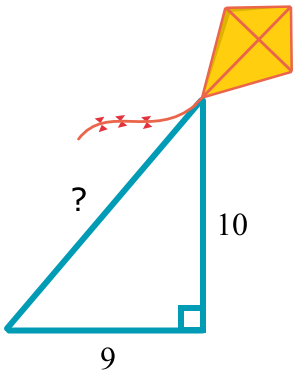
For the following right triangle, find the side length x. Round your answer to the nearest hundredth.





Question 6 of 50

A kite is flying 10 ft off the ground. Its line is pulled taut and casts a 9-ft shadow. Find the length of the line. If necessary, round your answer to the nearest tenth.



_____ ft

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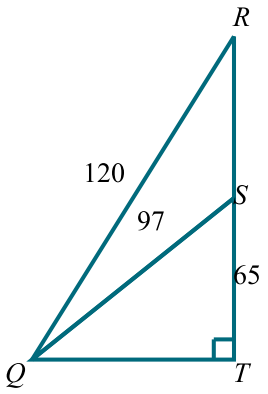
Question 7 of 50

Use the information given in the figure to find the length RT.
If applicable, round your answer to the nearest whole number.

The lengths on the figure are not drawn accurately.



Question 8 of 50

Determine whether a triangle with the given side lengths is a right triangle.

Side lengths	Right triangle	Not a right triangle	Not enough information
(a) 5, 11, 12	○	○	○
(b) 3, 9, 10	○	○	○
(c) 27, 36, 45	○	○	○
(d) 11, 13, 17	○	○	○

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Question 9 of 50

The side lengths of two triangles are given below.

Answer the questions about each triangle.

(a) One triangle has side lengths 24, 26, and 10. Compute the sum of the squares of the shorter lengths. $10^2 + 24^2 =$ _____ Compute the square of the longest length. $26^2 =$ _____ What kind of triangle is it? ○ Acute triangle ○ Right triangle ○ Obtuse triangle	**(b) The other triangle has side lengths 6, 11, and 8.** Compute the sum of the squares of the shorter lengths. $6^2 + 8^2 =$ _____ Compute the square of the longest length. $11^2 =$ _____ What kind of triangle is it? ○ Acute triangle ○ Right triangle ○ Obtuse triangle

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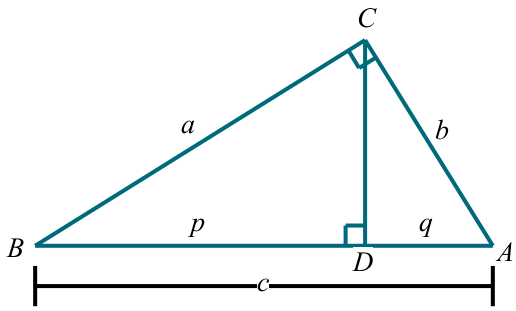
Question 10 of 50

In the figure below, $\triangle ADC \sim \triangle CDB \sim \triangle ACB$.



Fill in the following blanks using the lengths a, b, c, p, and q.

Part 1: Complete the proportions.

$\frac{c}{a} = \frac{a}{__}$ $\qquad$ $\frac{c}{b} = \frac{b}{__}$

Part 2: Use the method of cross products to rewrite the equations in Part 1.

$a^2 = c \cdot __$ $\qquad$ $b^2 = c \cdot __$

Part 3: Use Part 2 to fill in the blanks.

$a^2 + b^2 = c \cdot __ + c \cdot __$

Part 4: Factor the right-hand side of Part 3.

$a^2 + b^2 = c(__ + __)$

Part 5: Use the Segment Addition Property.

$p + q = __$

Part 6: Use Part 5 to rewrite the equation in Part 4.

$a^2 + b^2 = c(__)$

Part 7: Simplify.

$a^2 + b^2 = __^2$

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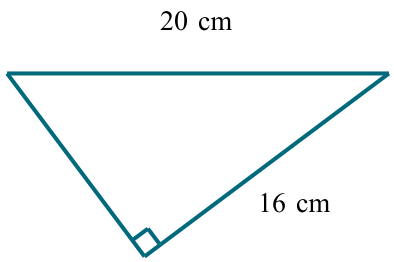
Question 11 of 50

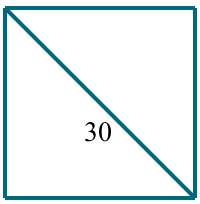
Find the area of the right triangle.
Be sure to include the correct unit in your answer.



Question 12 of 50

A square has a diagonal that is 30 yd long. What is the area of the square?

If necessary, round your answer to the nearest whole number.



Question 13 of 50

Simplify.

$$\frac{\sqrt{39}}{\sqrt{13}}$$

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Question 14 of 50

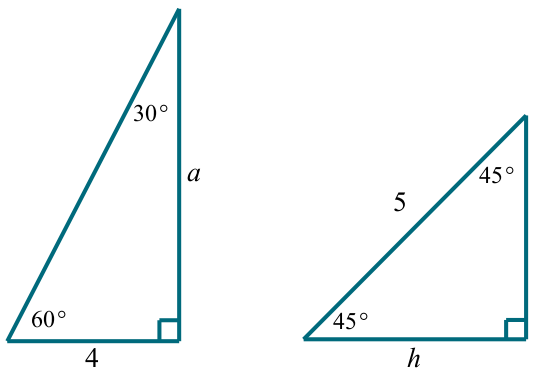
Rationalize the denominator and simplify.

$$\frac{8}{\sqrt{3}}$$

Question 15 of 50

For the right triangles below, find the values of the side lengths a and h.

Round your answers to the nearest tenth.



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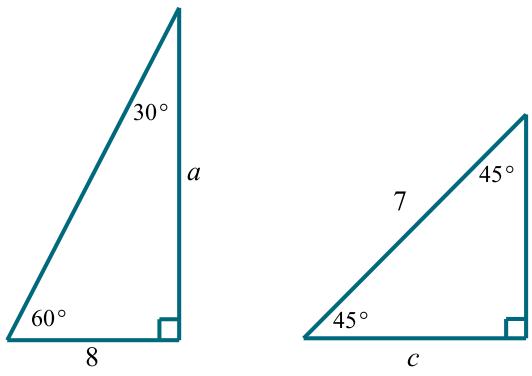
Question 16 of 50

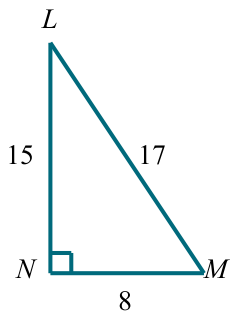
For the right triangles below, find the exact values of the side lengths a and c.

If necessary, write your responses in simplified radical form.



Question 17 of 50

A right triangle has side lengths 8, 15, and 17 as shown below.
Use these lengths to find $\tan M$, $\cos M$, and $\sin M$.



$\tan M =$ _____

$\cos M =$ _____

$\sin M =$ _____

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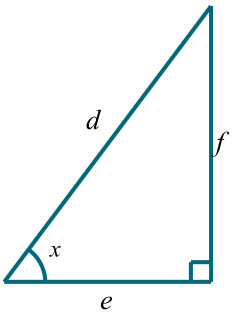
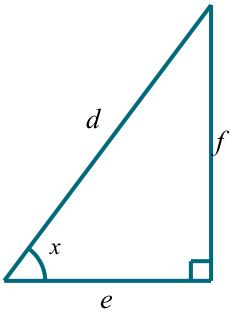
Question 18 of 50

A right triangle has side lengths d, e, and f as shown below.
Use these lengths to find $\tan x$, $\sin x$, and $\cos x$.





$\tan x =$ _____

$\sin x =$ _____

$\cos x =$ _____

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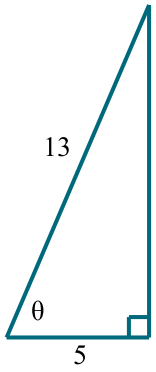
Question 19 of 50

Find $\cot\theta$, $\csc\theta$, and $\sin\theta$, where θ is the angle shown in the figure. Give exact values, not decimal approximations.

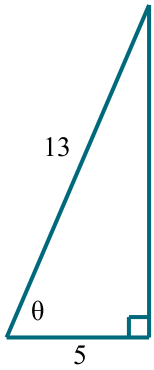


$\cot\theta =$ ☐

$\csc\theta =$ ☐

$\sin\theta =$ ☐

Find $\cot\theta$, $\csc\theta$, and $\sin\theta$, where θ is the angle shown in the figure. Give exact values, not decimal approximations.



$\cot\theta =$ _____

$\csc\theta =$ _____

$\sin\theta =$ _____

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Question 20 of 50

Use a calculator to evaluate each expression.
Round your answers to the nearest hundredth.

$\sin 11° =$ _____

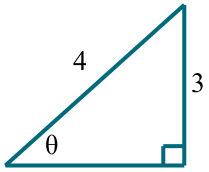
$\tan 89° =$ _____

$\cos 56° =$ _____

Question 21 of 50

Find $\cos\theta$, where θ is the angle shown.

Give an exact value, not a decimal approximation.



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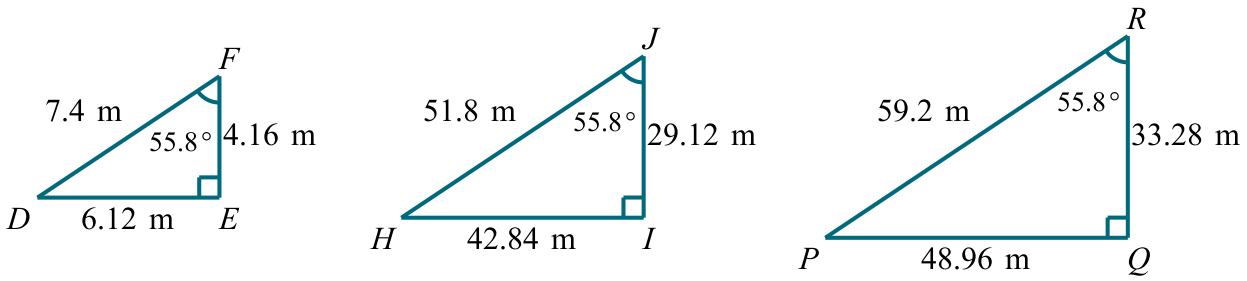
Question 22 of 50

The three right triangles below are similar.
The acute angles $\angle F$, $\angle J$, and $\angle R$ are all approximately measured to be $55.8°$.
The side lengths for each triangle are as follows.
Note that the triangles are not drawn to scale.



(a) For each triangle, find the ratio of the length of the side opposite $55.8°$ to the length of the hypotenuse. Round your answers to the nearest hundredth.

$\frac{DE}{DF} =$ $\frac{HI}{HJ} =$ $\frac{PQ}{PR} =$

(b) Use the ALEKS Calculator to find $\sin 55.8°$, $\cos 55.8°$, and $\tan 55.8°$.
Round your answers to the nearest hundredth.

$\sin 55.8° =$ $\cos 55.8° =$ $\tan 55.8° =$

(c) Which trigonometric function gives each ratio of sides in part (a)?

- sine
- cosine
- tangent
- none of these

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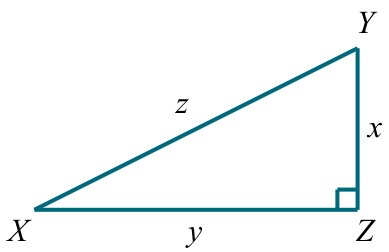
Question 23 of 50

The figure below is a right triangle with side lengths x, y, and z.

Suppose that $m\angle X$ does not equal $m\angle Y$.



Complete the following.

Part 1: Use x, y, and z to fill in the blanks.
Make sure to use the appropriate upper-case or lower-case letters.

$\sin X = \frac{\square}{\square}$ $\sin Y = \frac{\square}{\square}$

$\cos X = \frac{\square}{\square}$ $\cos Y = \frac{\square}{\square}$

Part 2: In $\triangle XYZ$, $\angle X$ and $\angle Y$ are

- neither complementary nor supplementary.
- supplementary.
- complementary.

Part 3: Select *all* of the true statements.

- ☐ $\cos X = \sin Y$
- ☐ $\sin X = \sin Y$
- ☐ $\sin X = \cos Y$
- ☐ $\cos X = \cos Y$
- ☐ None of the above is true.

Part 4: Fill in the blank.

$\sin(36°) = \cos(_____°)$

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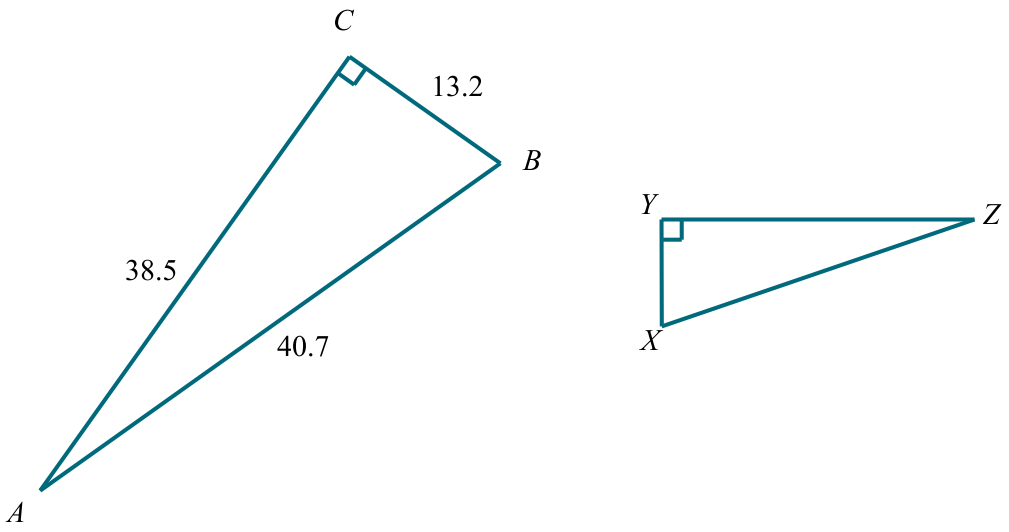
Question 24 of 50

In the figure below, $\triangle ABC \sim \triangle ZXY$.
Find $\sin Z$, $\tan Z$, and $\cos Z$.
Round your answers to the nearest hundredth.



$\sin Z =$ ☐

$\tan Z =$ ☐

$\cos Z =$ ☐

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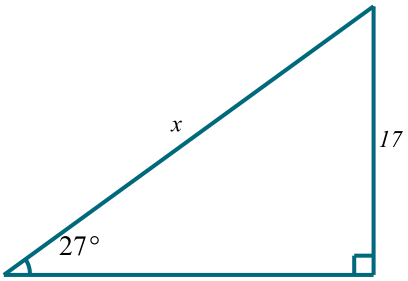
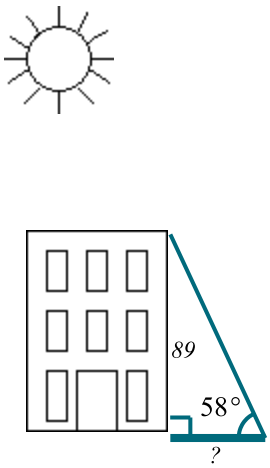
Question 25 of 50

Solve for x in the triangle. Round your answer to the nearest tenth.



Question 26 of 50

If the sun is $58°$ above the horizon, find the length of the shadow cast by a building 89 ft tall. Round your answer to the nearest tenth.



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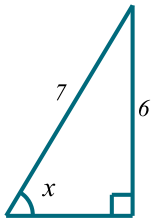
Question 27 of 50

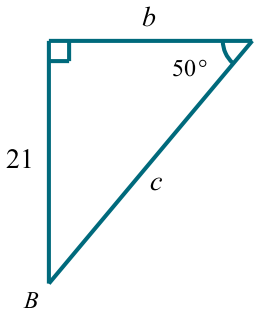
Find x. Round your answer to the nearest tenth of a degree.



Question 28 of 50

Solve the right triangle.

Round your answers to the nearest tenth.



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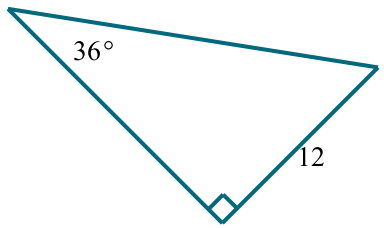
Question 29 of 50

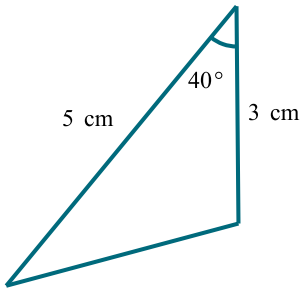
Find the area of the given triangle.
Round your answer to the nearest tenth. Do not round any intermediate computations.



Question 30 of 50

Find the area of the triangle below.

Carry your intermediate computations to at least four decimal places. Round your answer to the nearest hundredth.



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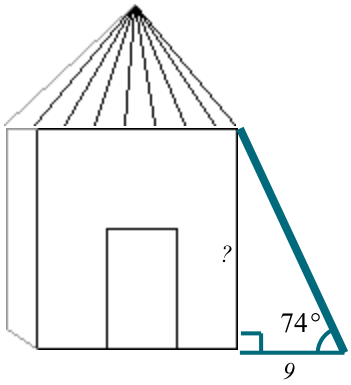
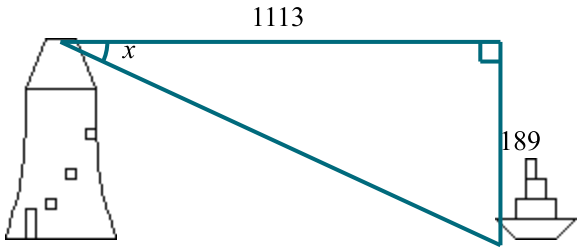
Question 31 of 50

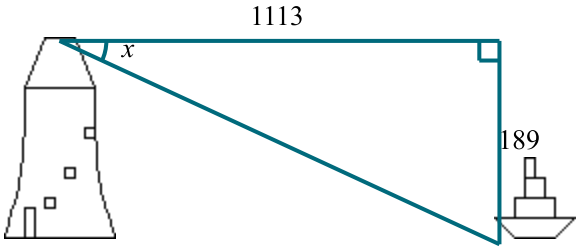
A ladder leans against the side of a house. The angle of elevation of the ladder is $74°$ when the bottom of the ladder is 9 ft from the side of the house. How high is the top of the ladder from the ground? Round your answer to the nearest tenth.



Question 32 of 50

Find x, the angle of depression from the top of a lighthouse that is 189 ft above water level to the waterline of a ship 1113 ft off shore. Round your answer to the nearest tenth of a degree.





$x = \square\,°$

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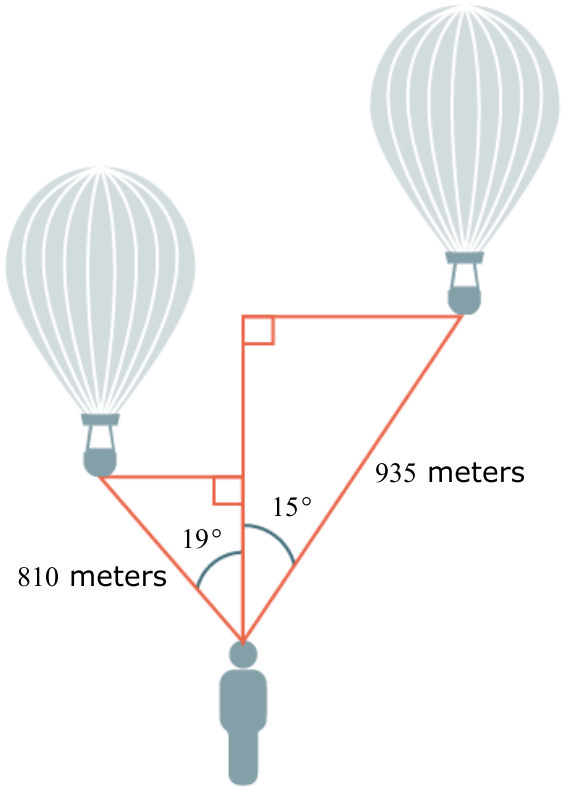
Question 33 of 50

Looking up, Heather sees two hot air balloons in the sky as shown. She determines that the lower hot air balloon is 810 meters away, at an angle of $19°$ from the vertical. The higher hot air balloon is 935 meters away, at an angle of $15°$ from the vertical. How much higher is the balloon on the right than the balloon on the left?

Do not round any intermediate computations. Round your answer to the nearest tenth.

Note that the figure below is not drawn to scale.



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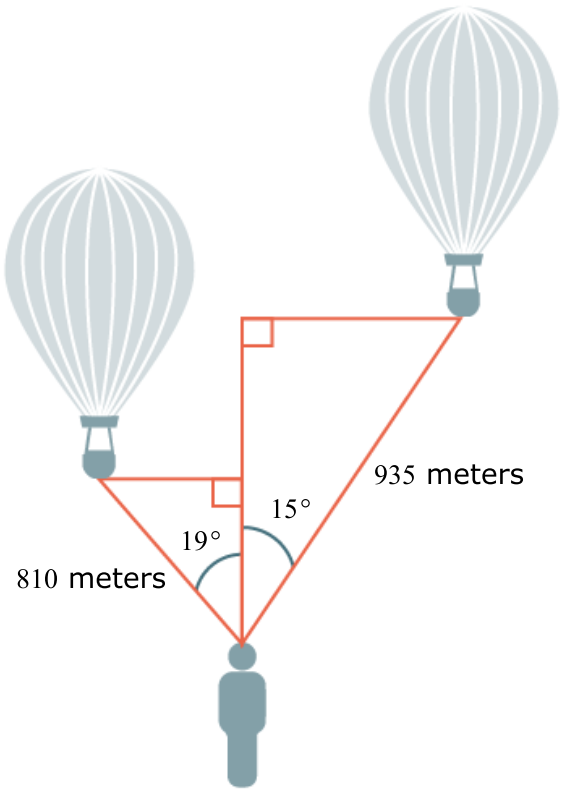
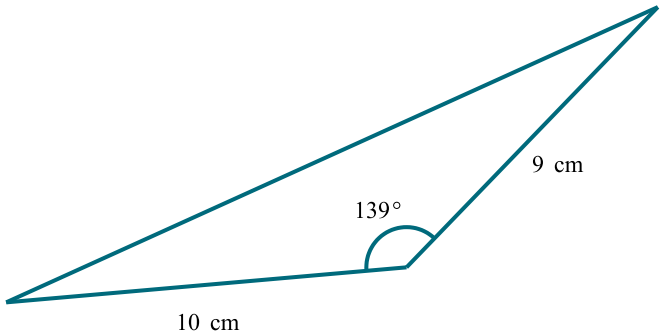


_____ meters

Question 34 of 50

Find the area of the triangle below.

Carry your intermediate computations to at least four decimal places. Round your answer to the nearest hundredth.



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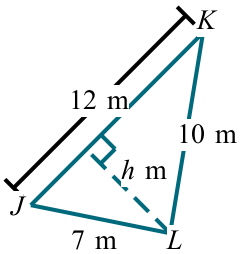
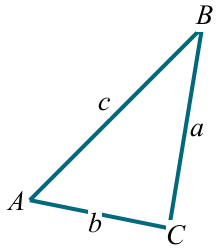
Question 35 of 50

Use $\triangle JKL$, as needed, to complete the given parts.



Part 1:	Let h be the height of $\triangle JKL$. Find the length of the corresponding base. _____ m
Part 2:	The measure of angle J is $56.39°$. Fill in the blank to complete the sine ratio. $\sin(56.39°) = \frac{h}{___}$
Part 3:	Fill in the blanks in the expression for the area of $\triangle JKL$. Area of $\triangle JKL = \frac{1}{2} \cdot ___ \cdot ___ \cdot \sin(56.39°)\ \text{m}^2$
Part 4:	Write the area of $\triangle ABC$ in terms of b, c, and A.  Area of $\triangle ABC =$ _____

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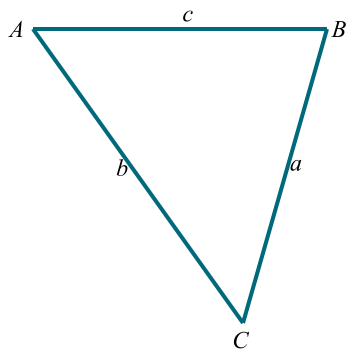
Question 36 of 50

Consider a triangle ABC like the one below. Suppose that $A = 83°$, $B = 61°$, and $c = 25$. (The figure is not drawn to scale.) Solve the triangle.

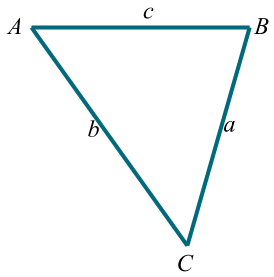
Round your answers to the nearest tenth.



Question 37 of 50

Consider a triangle ABC like the one below. Suppose that $b = 36$, $a = 41$, and $B = 40°$. (The figure is not drawn to scale.) Solve the triangle.

Carry your intermediate computations to at least four decimal places, and round your answers to the nearest tenth.



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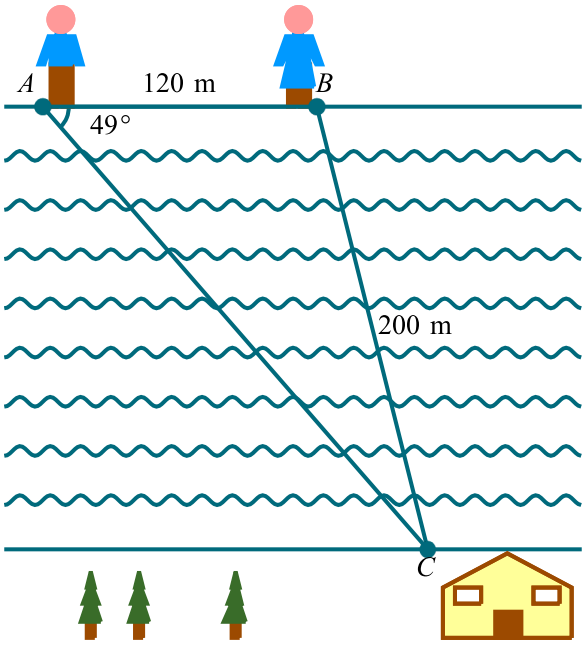
Question 38 of 50

Austin and Keisha are standing on a riverbank, 120 meters apart, at points A and B respectively. (See the figure below.) Keisha is 200 meters from a house located across the river at point C. Suppose that angle A (angle BAC) is $49°$. What is the measure of angle B (angle ABC)? Round your answer to the nearest tenth of a degree.



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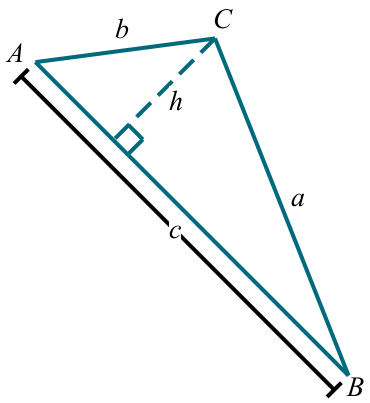
Question 39 of 50

The side lengths for $\triangle ABC$ below are a, b, and c.
The height of $\triangle ABC$ is h.

Fill in the blanks using the lengths a, b, c, and h to derive the Law of Sines.
Make sure to use the appropriate upper-case or lower-case letters.



Part 1: Use trigonometry to fill in the blanks.

$\sin A =$ ___ and $\sin B =$ ___

Part 2: Rewrite the equations from Part 1.

h = ___ $\cdot \sin A$ and h = ___ $\cdot \sin B$

Part 3: Use the equations from Part 2 to write an equation relating $\sin A$ and $\sin B$.

___ $\cdot \sin A$ = ___ $\cdot \sin B$

Part 4: Rewrite the equation from Part 3.

$$\frac{\sin A}{__} = \frac{\sin B}{__}$$

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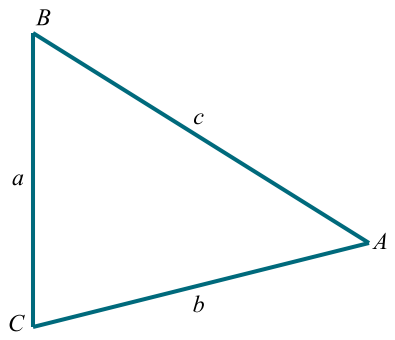
Question 40 of 50

Consider a triangle ABC like the one below. Suppose that $B = 29°$, $C = 115°$, and $a = 73$. (The figure is not drawn to scale.) Solve the triangle.

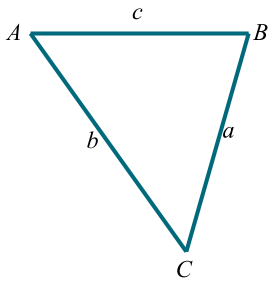
Round your answers to the nearest tenth.



Question 41 of 50

Consider a triangle ABC like the one below. Suppose that $c = 12$, $a = 19$, and $C = 73°$. (The figure is not drawn to scale.) Solve the triangle.

Carry your intermediate computations to at least four decimal places, and round your answers to the nearest tenth.



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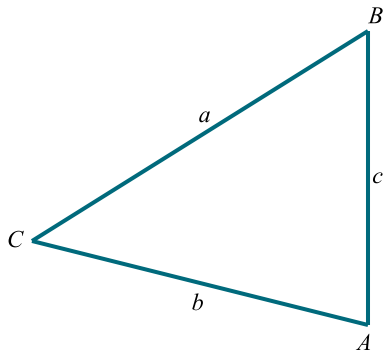
Question 42 of 50

Consider a triangle ABC like the one below. Suppose that $a = 34$, $b = 37$, and $c = 41$. (The figure is not drawn to scale.) Solve the triangle.

Carry your intermediate computations to at least four decimal places, and round your answers to the nearest tenth.



$A = \square°$, $B = \square°$, $C = \square°$

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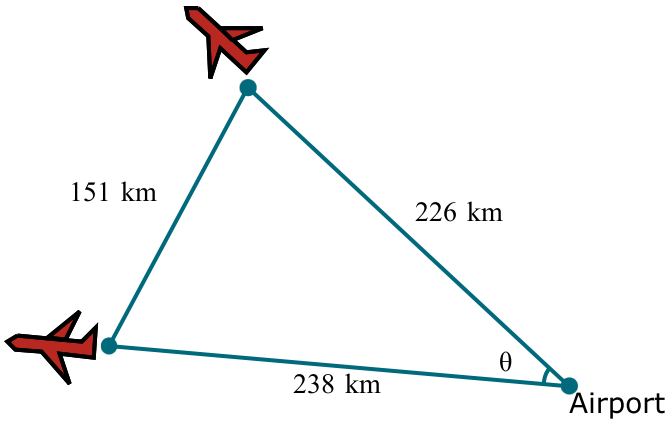
Question 43 of 50

Two airplanes leave an airport at the same time. An hour later, the planes are $151\ \text{km}$ apart. If one plane has traveled $238\ \text{km}$ and the other has traveled $226\ \text{km}$ during the hour, find the angle θ between their flight paths. See the figure below.



Carry your intermediate computations to at least four decimal places.
Round your answer to the nearest tenth of a degree.

Question 44 of 50

Solve the system of equations.

$$y = 6x + 21$$

$$y = 2x + 37$$

Question 45 of 50

Simplify.

$$y^5 \cdot y \cdot y^2$$

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Question 46 of 50

Simplify.

$$\sqrt{32} \cdot \sqrt{2}$$

Question 47 of 50

Simplify.

$$\sqrt{80} \times 4\sqrt{20}$$

Question 48 of 50

Simplify.

$$\sqrt{6} \cdot \sqrt{5}$$

Question 49 of 50

Simplify.

$$\sqrt{6} \cdot \sqrt{3}$$

Question 50 of 50

Simplify.

$$\sqrt{12} \times 4\sqrt{27}$$

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **50**

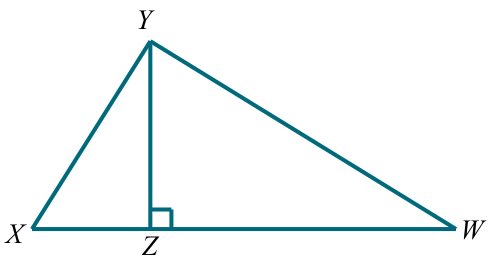
Question 1 of 50

$HN = 122$

$LJ = 120$

$GJ = 164$

Question 2 of 50



(a) Write a similiarity statement relating the three right triangles.

$\triangle YWZ \sim \triangle XYZ \sim \triangle XWY$

(b) Complete each proportion.

$\frac{XZ}{XY} = \frac{XY}{XW}$ $\frac{WZ}{YZ} = \frac{YZ}{XZ}$

Question 3 of 50

$y = \sqrt{12}$

Question 4 of 50

41

Question 5 of 50

12

Question 6 of 50

13.5 ft

Question 7 of 50

96

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Question 8 of 50

Side lengths	Right triangle	Not a right triangle	Not enough information
(a) 5, 11, 12	○	◉	○
(b) 3, 9, 10	○	◉	○
(c) 27, 36, 45	◉	○	○
(d) 11, 13, 17	○	◉	○

Question 9 of 50

(a) One triangle has side lengths 24, 26, and 10.

Compute the sum of the squares of the shorter lengths.

$$10^2 + 24^2 = 676$$

Compute the square of the longest length.

$$26^2 = 676$$

What kind of triangle is it?

- ○ Acute triangle
- ◉ Right triangle
- ○ Obtuse triangle

(b) The other triangle has side lengths 6, 11, and 8.

Compute the sum of the squares of the shorter lengths.

$$6^2 + 8^2 = 100$$

Compute the square of the longest length.

$$11^2 = 121$$

What kind of triangle is it?

- ○ Acute triangle
- ○ Right triangle
- ◉ Obtuse triangle

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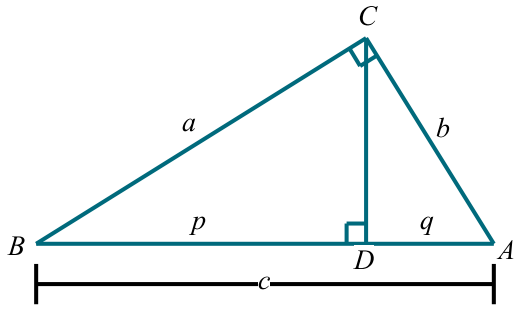
Question 10 of 50



Fill in the following blanks using the lengths a, b, c, p, and q.

Part 1: Complete the proportions.

$$\frac{c}{a} = \frac{a}{p} \qquad \frac{c}{b} = \frac{b}{q}$$

Part 2: Use the method of cross products to rewrite the equations in Part 1.

$$a^2 = c \cdot p \qquad b^2 = c \cdot q$$

Part 3: Use Part 2 to fill in the blanks.

$$a^2 + b^2 = c \cdot p + c \cdot q$$

Part 4: Factor the right-hand side of Part 3.

$$a^2 + b^2 = c(p + q)$$

Part 5: Use the Segment Addition Property.

$$p + q = c$$

Part 6: Use Part 5 to rewrite the equation in Part 4.

$$a^2 + b^2 = c(c)$$

Part 7: Simplify.

$$a^2 + b^2 = c^2$$

Question 11 of 50

Area: 96 cm^2

Question 12 of 50

450 yd^2

Question 13 of 50

$\sqrt{3}$

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Question 14 of 50

$$\frac{8\sqrt{3}}{3}$$

Question 15 of 50

(a) $a = 6.9$
(b) $h = 3.5$

Question 16 of 50

$a = 8\sqrt{3}$
$c = \frac{7\sqrt{2}}{2}$

Question 17 of 50

$$\tan M = \frac{15}{8}$$

$$\cos M = \frac{8}{17}$$

$$\sin M = \frac{15}{17}$$

Question 18 of 50

$$\tan x = \frac{f}{e}$$

$$\sin x = \frac{f}{d}$$

$$\cos x = \frac{e}{d}$$

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Question 19 of 50

$$\cot\theta = \frac{5}{12}$$

$$\csc\theta = \frac{13}{12}$$

$$\sin\theta = \frac{12}{13}$$

Question 20 of 50

$\sin 11° = 0.19$

$\tan 89° = 57.29$

$\cos 56° = 0.56$

Question 21 of 50

$$\cos\theta = \frac{\sqrt{7}}{4}$$

Question 22 of 50

(a) $\frac{DE}{DF} = 0.83$ $\frac{HI}{HJ} = 0.83$ $\frac{PQ}{PR} = 0.83$

(b) $\sin 55.8° = 0.83$ $\cos 55.8° = 0.56$ $\tan 55.8° = 1.47$

(c)
- (●) sine
- ( ) cosine
- ( ) tangent
- ( ) none of these

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Question 23 of 50

Part 1: Use x, y, and z to fill in the blanks.
Make sure to use the appropriate upper-case or lower-case letters.

$\sin X = \frac{x}{z}$ $\sin Y = \frac{y}{z}$

$\cos X = \frac{y}{z}$ $\cos Y = \frac{x}{z}$

Part 2: In $\triangle XYZ$, $\angle X$ and $\angle Y$ are

- complementary.

Part 3: Select *all* of the true statements.

- [x] $\cos X = \sin Y$
- [ ] $\sin X = \sin Y$
- [x] $\sin X = \cos Y$
- [ ] $\cos X = \cos Y$
- [ ] None of the above is true.

Part 4: Fill in the blank.

$\sin(36°) = \cos(54°)$

Question 24 of 50

$\sin Z = 0.32$

$\tan Z = 0.34$

$\cos Z = 0.95$

Question 25 of 50

$x = 37.4$

Question 26 of 50

55.6ft

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Question 27 of 50

$x = 59.0°$

Question 28 of 50

$B = 40°$
$b = 17.6$
$c = 27.4$

Question 29 of 50

99.1 square units

Question 30 of 50

4.82 cm^2

Question 31 of 50

31.4ft

Question 32 of 50

$x = 9.6°$

Question 33 of 50

137.3 meters

Question 34 of 50

29.52 cm^2

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Question 35 of 50

Part 1: Let h be the height of $\triangle JKL$.

Find the length of the corresponding base.

12 m

Part 2: The measure of angle J is $56.39°$.

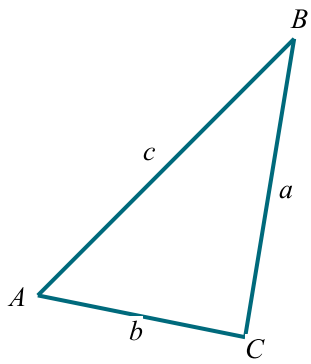
Fill in the blank to complete the sine ratio.

$\sin(56.39°) = \frac{h}{7}$

Part 3: Fill in the blanks in the expression for the area of $\triangle JKL$.

Area of $\triangle JKL = \frac{1}{2} \cdot 12 \cdot 7 \cdot \sin(56.39°)$ m^2

Part 4: Write the area of $\triangle ABC$ in terms of b, c, and A.



Area of $\triangle ABC = \frac{1}{2} cb \sin A$

Question 36 of 50

$C = 36°$, $a = 42.2$, $b = 37.2$

Question 37 of 50

$A = 47.1°$, $C = 92.9°$, $c = 55.9$,

or $A = 132.9°$, $C = 7.1°$, $c = 6.9$

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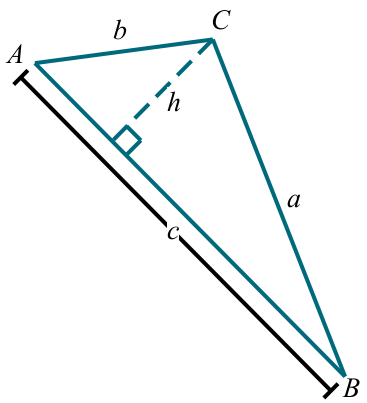
Question 38 of 50

$104.1°$

Question 39 of 50



Part 1: Use trigonometry to fill in the blanks.

$\sin A = \frac{h}{b}$ and $\sin B = \frac{h}{a}$

Part 2: Rewrite the equations from Part 1.

$h = b \cdot \sin A$ and $h = a \cdot \sin B$

Part 3: Use the equations from Part 2 to write an equation relating $\sin A$ and $\sin B$.

$b \cdot \sin A = a \cdot \sin B$

Part 4: Rewrite the equation from Part 3.

$\frac{\sin A}{a} = \frac{\sin B}{b}$

Question 40 of 50

$A = 36°,\ b = 60.2,\ c = 112.6$

Question 41 of 50

No solution

Question 42 of 50

$A = 51.4°,\ B = 58.2°,\ C = 70.4°$

Question 43 of 50

$37.9°$

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Question 44 of 50

$x = 4$
$y = 45$

Question 45 of 50

y^8

Question 46 of 50

8

Question 47 of 50

160

Question 48 of 50

$\sqrt{30}$

Question 49 of 50

$3\sqrt{2}$

Question 50 of 50

72

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